

Dataset generation for chiral metamaterials (notes)

Contents

1 Unit cell based generation

1.1 Nassar-type unit cells.

Nassar et al. [3] design a lattice unit cell for elastic cloaking that exhibits a rank-3 elasticity tensor *lacking minor symmetry*, as required by the Brun–Guenneau–Movchan (BGM) gauge of transformation elasticity [1]. The same unit-cell family is directly relevant here because it provides a parametric, analytically tractable class of *chiral* (polar) lattices whose effective properties can be sampled systematically.

Unit-cell architecture. A rectangular unit cell with aspect ratio a/b contains:

- two diagonal springs of stiffness α , inclined at angle θ to the $\mathbf{n}$ -direction;
- one vertical spring of stiffness β ;
- a single rigid mass m subjected to a restoring torque of stiffness κ (torque per radian per unit cell area).

All contacts between the mass and the lattice frame are hinge-like, so the cell supports no couple stresses from the springs alone. The restoring torque κ is what makes the lattice *polar*: it generates asymmetric stress and breaks the minor symmetry of the effective elasticity tensor.

Effective constitutive law. In the normalised polar basis $(\mathbf{m}, \mathbf{n})$, the homogenised Hooke’s law reads

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \\ \sigma_{21} \end{bmatrix} = \begin{bmatrix} \frac{ac^2\alpha}{b} & cs\alpha & 0 & 0 \\ cs\alpha & \frac{b(as^2\alpha + 2\beta)}{a} & 0 & 0 \\ 0 & 0 & \frac{b^2c^2\alpha\kappa}{\alpha(bc-as)^2 + ab\kappa} & \frac{abcs\alpha\kappa}{\alpha(bc-as)^2 + ab\kappa} \\ 0 & 0 & \frac{abcs\alpha\kappa}{\alpha(bc-as)^2 + ab\kappa} & \frac{a^2s^2\alpha\kappa}{\alpha(bc-as)^2 + ab\kappa} \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{22} \\ e_{12} \\ e_{21} \end{bmatrix}, \quad (1)$$

where $c = \cos \theta$, $s = \sin \theta$. The off-diagonal shear block is symmetric in the 12/21 sense (both off-diagonal entries equal), while the normal block is symmetric; the tensor therefore has major symmetry $C_{ijkl} = C_{klij}$ but lacks minor symmetry $C_{ijkl} \neq C_{jikl}$ in general.

Inverse identification from an augmented Voigt matrix. Conversely, given an arbitrary augmented Voigt stiffness matrix C_{IJ} (indices $I, J \in \{11, 22, 12, 21\}$) that is block-diagonal and whose shear block is rank-1, one can recover the five unit-cell parameters in closed form by matching entries of (??).

Eliminating the unknowns sequentially yields:

$$\theta = \arctan \sqrt{\frac{C_{12,21} C_{11,22}}{C_{11,11} C_{12,12}}}, \quad (2)$$

$$\alpha = \frac{C_{11,22}}{cs}, \quad (3)$$

$$\frac{a}{b} = \frac{s C_{11,11}}{c C_{11,22}}, \quad (4)$$

$$\beta = \frac{1}{2} \left(\frac{a}{b} C_{22,22} - \alpha s^2 \right), \quad (5)$$

$$\kappa = \frac{C_{12,12} \alpha (c - rs)^2}{c^2 \alpha - C_{12,12} r}. \quad (6)$$

The rank-1 condition $C_{12,12} C_{21,21} = C_{12,21}^2$ serves as a consistency check: it is satisfied identically for any stiffness produced by (??). When $c^2 \alpha = C_{12,12} r$ the denominator in (??) vanishes and $\kappa \rightarrow \infty$, corresponding to the minor-symmetric limit $\lambda = \mu$.

Dataset generation strategy. A parametric dataset of Nassar-type unit cells can be generated by sampling the five parameters $(\lambda, \mu, f, \theta, \kappa)$, where f is a normalised radial position in the cloak that controls the aspect ratio a/b via (??). The ground-truth labels can be computed either analytically from (??) or numerically via FE homogenisation.

1.2 Cellular automaton based generation.

Nakarmi et al. [2] propose a data-driven framework for predicting the non-linear stress-strain response of cellular structures under large uniaxial compression. The key novelty relevant to dataset generation is their use of *cellular automata* (CA) to generate a large and topologically diverse library of unit cells.

Grid and initial state. Each unit cell is discretised on a 25×25 pixel grid (physical size $2.5 \text{ mm} \times 2.5 \text{ mm}$). Each pixel takes a binary value: 1 (solid material, *live*) or 0 (void, *dead*). The initial population of live pixels is approximately 49% (≈ 306 out of 625).

Evolution rules. The topology evolves according to the following rules, applied simultaneously to all interior pixels at every iteration:

1. *Border enforcement:* border pixels are permanently set to live, except for a random subset of adjacent *gate* pixels that are set to dead, ensuring boundary connectivity while allowing some porosity.
2. *Birth:* a dead pixel becomes live if it has 5–8 live neighbours.
3. *Death:* a live pixel becomes dead if it has 0–3 live neighbours.
4. *Survival:* a pixel retains its current value if it has exactly 4 live neighbours.

Iteration continues until convergence (typically 7 steps with no further change).

Post-processing for connectivity. The converged CA map is inverted to yield a *reverse map* (the solid phase). A flood-fill algorithm identifies all disconnected solid regions $\mathcal{R}$. A connection graph $G(u, v, m, n, r)$ is then built, where $u, v \in \mathcal{R}$ are pairs of regions, m, n are the nearest cells between them, and r is the minimum distance. Kruskal’s minimum spanning tree algorithm on G produces a spanning graph K that links every disconnected region with minimum total bridge length; bridges are drawn as 4-pixel-wide connections. Finally, in the original paper, the structure is mirrored along the vertical and horizontal axes to produce a periodic unit cell.

Chiral assembly. The original CA procedure produces unit cells that are symmetric under reflection. To introduce chirality, we can assemble four rotated copies of a single 25×25 CA tile — at 0° , 90° , 180° , and 270° — into a single 50×50 unit cell (Figure ??). Gate pixels are centred on each tile edge so that, after rotation, opposing gates remain aligned and the assembled cell stays connected across the internal seams. Figure ?? shows successive stages of this process from a freshly evolved CA map through to the final chiral 50×50 cell.

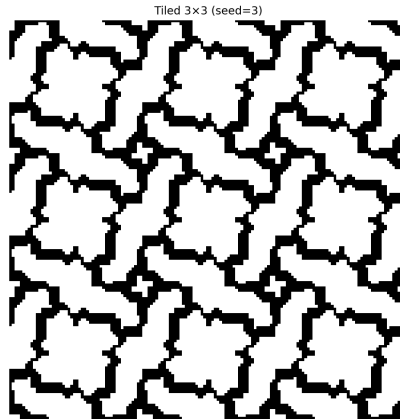


Figure 1: A single CA tile 3x3 obtained by assembling unit cells.

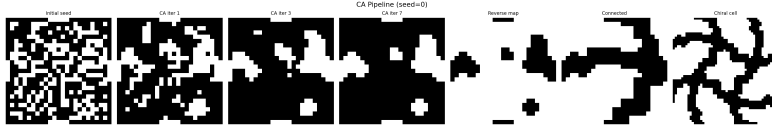


Figure 2: Progression from random initialization to final chiral unit cell across the stages.

Squared (D_2) assembly. For bulk dataset generation we also use a non-chiral variant in which the four copies of the 25×25 CA tile are *mirrored* rather than rotated (the `squared` mode of `dataset/cellular_chiral/bulk_generate.py`, run as `python -m dataset.cellular_chiral.bulk_generate --assembly squared`). The top-left quadrant is copied directly, the top-right is reflected left-right, the bottom-left up-down, and the bottom-right across both axes, so the assembled 50×50 cell is invariant under reflection about both the horizontal and vertical centre lines: it possesses D_2 (orthotropic) symmetry. Each cell is generated with its live fraction drawn uniformly in $[0.20, 0.80]$, and the whole library is written contiguously to a single memmapped array. The D_2 mirror symmetry forces the effective stiffness to be orthotropic in the cell axes: the normal-shear coupling components vanish ($C_{1112} = C_{2212} = 0$) and minor symmetry is retained, so the Cosserat shear asymmetry of (??) disappears ($C_{1212} = C_{2121}$, hence the chirality index $\chi = 0$). The tensor is then fully described by only four independent components

$$(C_{1111}, C_{2222}, C_{1212}, C_{1122}), \quad (7)$$

which is exactly the *flat4* parametrisation implemented by `C.to_flatC/flatC.to.C` in `rayleigh_cloak/materials.py`. This compact 4-vector (together with the density ρ) is the per-cell label stored for the squared dataset and the feature space over which the dataset GMM is fitted. The chiral cells of the previous paragraph break minor symmetry and instead require the 6- or 10-parameter Cosserat representations; the *flat4* form is appropriate precisely because the squared cells are minor-symmetric.

2 Metrics of goodness of generated dataset

Diversity. Diversity measures how spread out the dataset is in feature space. In the Nakarmi et al. framework [2], diversity of the stress-strain curves is assessed via the explained variance captured by the first few PCA components: a dataset in which 99.89% of variance is captured by 20 components is well-concentrated along those principal directions.

Coverage. Coverage measures how well the dataset fills the target design space — i.e. whether all practically relevant property combinations are represented. It can be quantified as the fraction of a reference grid in property space

that contains at least one sample within a radius ε , or via the Hausdorff distance between the sample set and the target region.

Chirality. For metamaterial cloaking applications, the dataset must contain a sufficient fraction of chiral unit cells (those with broken minor symmetry). Chirality can be measured by the asymmetry index χ defined above:

$$\chi = \frac{|C_{1212} - C_{2121}|}{C_{1212} + C_{2121}}, \quad (8)$$

References

- [1] M. Brun, S. Guenneau, and A. B. Movchan. “Achieving control of in-plane elastic waves”. In: *Applied Physics Letters* 94.6 (2009), p. 061903. DOI: 10.1063/1.3068491.
- [2] Sushan Nakarmi et al. “Predicting non-linear stress–strain response of mesostructured cellular materials using supervised autoencoder”. In: *Computer Methods in Applied Mechanics and Engineering* 432 (2024), p. 117372. DOI: 10.1016/j.cma.2024.117372.
- [3] H. Nassar, Y. Y. Chen, and G. L. Huang. “A degenerate polar lattice for cloaking in full two-dimensional elastodynamics and statics”. In: *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 474.2218 (2018), p. 20180523. DOI: 10.1098/rspa.2018.0523.